General Relativity Final Examination — Marking Summary

Each script in `runs/` was marked against `solutions.tex`. The paper has five questions worth 20 points in total (Q1: 2; Q2: 2+1; Q3: 2+2; Q4: 2+1+2; Q5: 1+2+3). Half marks were allowed.

		Q1	**Q2**		**Q3**		**Q4**			**Q5**			**Total**
Model	**Effort**	(2)	a (2)	b (1)	a (2)	b (2)	a (2)	b (1)	c (2)	a (1)	b (2)	c (3)	**(20)**
claude-haiku-5-5	low	2	2	1	2	2	2	1	1.5	1	2	3	**19.5**
claude-haiku-5-5	high	2	2	1	2	2	2	1	2	1	2	3	**20**
claude-sonnet-5-5	low	2	2	1	2	2	2	1	2	1	2	3	**20**
claude-sonnet-5-5	high	2	2	1	2	2	2	1	2	1	2	3	**20**
claude-opus-5-5	low	2	2	1	2	2	2	1	2	1	2	3	**20**
claude-opus-5-5	high	2	2	1	2	2	2	1	2	1	2	3	**20**
gemini-3.8-flash	low	2	2	1	2	2	2	1	2	1	2	3	**20**
gemini-3.8-flash	high	2	2	1	2	2	2	1	2	1	2	3	**20**
gemini-4-argon	low	2	2	1	2	2	2	1	2	1	2	3	**20**
gemini-4-argon	high	2	2	1	2	2	2	1	2	1	2	3	**20**

Comments on individual scripts

claude-haiku-5-5 (low), 19.5/20. All derivations are correct and complete. Q4c loses half a mark: the conformal metric and the square Penrose diagram are right, but the light rays drawn in the TikZ figure have arrows running from the top corners ($\mathcal{I}^+$) down to the centre. These are past-directed, and they contradict the accompanying text, which says light rays from the poles reach $\mathcal{I}^+$.

claude-haiku-5-5 (high), 20/20. Complete and correct. It derives the Kretschmann scalar from orthonormal-frame Riemann components, checks the Friedmann equations, and gives a quadrupole-formula inspiral estimate.

claude-sonnet-5-5 (low), 20/20. Concise but every required step is present. The Penrose diagram is correct, with horizons, and Q5c connects chirp mass, distance, total mass and remnant spin to the signal.

claude-sonnet-5-5 (high), 20/20. Complete and correct, with a detailed discussion of horizons in the de Sitter Penrose diagram and of interferometric detection.

claude-opus-5-5 (low), 20/20. Complete and correct. It includes the differential light-travel time $\Delta T \approx 2h_+ L/c$ and the extraction of the binary's parameters from the signal.

claude-opus-5-5 (high), 20/20. Complete and correct throughout. It analyses null rays in Eddington–Finkelstein coordinates and identifies the static patch in the Penrose diagram, and its pgfplots chirp waveform is physically motivated.

gemini-3.8-flash (low), 20/20. Complete and correct. It shows that light cones are preserved (Q4b), gives a labelled Penrose diagram and a waveform sketch. The link between signal and binary physics ($f_{\rm GW} = 2f_{\rm orb}$, ISCO, quasi-normal modes giving M_f, a_f) is adequate.

gemini-3.8-flash (high), 20/20. Complete and very detailed, with full step-by-step derivations. It includes the inverse EF metric, the infall proper time, the chirp-mass formula and standard sirens. (There is a cosmetic typo `\ partial` in the restated question; the answers are unaffected.)

gemini-4-argon (low), 20/20. Complete and correct. It adds the event-horizon and causal-trapping discussion in Q3, and shows how amplitude and frequency scale with orbital separation in Q5c.

gemini-4-argon (high), 20/20. Complete and correct, with comparable depth to the low-effort script and a clean Penrose diagram and waveform.

Marking notes

- The reference solutions contain a few slips. Scripts were marked on correct physics rather than on literal agreement with the key:
  - Q4a: the key gives the scale factor as $L^2\cosh^2(t/L)$. The correct value is $a(t) = L\cosh(t/L)$ ($a^2 = L^2\cosh^2$), and every script gave this.
  - Q2a: the key writes the antisymmetric part $\hat{\Gamma}_{[\mu\nu]}{}^{\rho} = \frac{1}{12}H_{\mu\nu}{}^{\rho}$. The torsion tensor $T = \hat{\Gamma}_{\mu\nu}{}^{\rho} - \hat{\Gamma}_{\nu\mu}{}^{\rho} = \frac{1}{6}H_{\mu\nu}{}^{\rho}$, which is what all scripts found. Both say "not torsion-free".
  - Q2b: the key's final line should read $\hat{\Gamma}_{(\nu\rho)}{}^{\mu} = \Gamma_{\nu\rho}{}^{\mu}$ (symmetric part), not $[\nu\rho]$.
  - Q4c: the key's range "from $\cos\eta = -1$ … to $\cos\eta = -1$" should read η from $-\pi/2$ to $+\pi/2$.
- All ten scripts reached the correct conclusions on every part: not torsion-free but metric compatible; $r = 0$ is a curvature singularity and $r = 2GM$ a coordinate singularity; the ingoing EF metric; $a(t) = L\cosh(t/L)$; a square Penrose diagram with spacelike $\mathcal{I}^{\pm}$; Lorenz gauge leading to $\Box\bar{h}_{\mu\nu} = -16\pi G T_{\mu\nu}$; and $\Delta s \approx L(1 \pm \frac{1}{2}h_{+})$. Every script also included a sketch of the chirp signal.
- `summary.csv`: grades are given for the ten rows of the current run grid, which match the directories in `runs/`. The first four rows come from earlier `claude-haiku-5-5` launches, including two with run index 2. Their outputs are no longer on disk because later runs overwrote them, so their grade field is left blank.